

# **AI-GENERATED MODEL PAPER**

Subject: Database Management Systems

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*Total Marks: 100*

# Examination Paper

## Question Q1 (20 marks)

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- a) [PLACEHOLDER FIGURE] (A University Library ER diagram showing relationships between students, books, sections, librarians and members) (5)
- b) Explain the concept of aggregation in the context of the University Library scenario. How does it relate to the ternary relationship and what are its advantages? (5)
- c) Describe how you would enforce overlapping constraints in the database schema for the library system. Provide examples of how this would impact the data. (5)
- d) Calculate the number of unique combinations of book titles and authors. Use a mathematical formula to determine this value. (3)
- e) [PLACEHOLDER TABLE] (A table designed to store member information, including membership number, address, phone number, and borrowing history) (3)

## Question Q2 (20 marks)

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- a) Define the concept of normalization in database systems. (5)
- b) Explain the difference between partial and full functional dependency. (1)
- c) [PLACEHOLDER FIGURE] (Draw an ER diagram for the Library Book Borrowing System) (2)
- d) Identify the key attributes for the relation. (4)
- e) Normalize the relation to 2NF and explain the reasoning. (2)
- f) Define and explain the key concepts related to frame bytes time. (2)
- g) Construct an ER/EER diagram for the scenario described above, showing all entities, relationships, and attributes. (4)

## Question Q3 (22 marks)

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- a) [PLACEHOLDER FIGURE] (Describe an ER diagram with entities Books, Borrowers, Librarians, and Borrowings) (3)

- b) Explain how the system can ensure that only authorized users can borrow books. Please provide a detailed explanation of your answer. (2)
- c) Design a trigger to prevent borrowers from exceeding the library's maximum withdrawal limit per day. Provide a SQL script for the trigger and explain its functionality. (3)
- d) [PLACEHOLDER FIGURE] (Map the ER diagram to relational model tables for Books, Borrowers, Librarians, and Borrowings) (4)
- e) Explain how the system can handle cases where multiple copies of the same book are borrowed simultaneously. Please provide a detailed explanation of your answer. (4)
- f) Describe the benefits of using a database management system in this context. Please provide a detailed explanation of your answer. (3)
- g) Discuss potential security measures to protect the library's data from unauthorized access. Please provide a detailed explanation of your answer. (3)

#### Question Q4 (20 marks)

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- a) Describe the key components of an Entity-Relationship (ER) diagram and explain their significance in designing a database schema. (3)
- b) Design an ER diagram for the University Library's cataloging system, including entities such as Books, Borrowers, Librarians, and Borrowings. Explain how the system can ensure that only authorized users can borrow books. (3)
- c) Explain the concept of normalization in database systems and provide an example of a poorly normalized schema. How would you normalize this schema to achieve data integrity? (1)
- d) Calculate the primary key for the Books entity in the University Library's cataloging system, considering the attributes 'Book ID', 'Title', and 'Author'. Explain why these attributes are chosen as the primary key. (3)
- e) Design a relational schema for the Books entity, including the primary key, foreign keys, and data types for each attribute. Ensure that the schema adheres to the principles of normalization. (3)
- f) Explain how the University Library's cataloging system can implement data integrity by enforcing referential constraints between entities. Provide an example of a foreign key relationship. (4)
- g) Describe the benefits of using a normalized database schema for the University Library's cataloging system, including improved data consistency and reduced data

### Question Q5 (18 marks)

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- a) Draw an ER diagram for the University Library's cataloging system, including entities Books, Borrowers, Librarians, and Borrowings. Explain how the system can ensure that only authorized users can borrow books. (6)
- b) [PLACEHOLDER FIGURE] (Describe the relationship between the Book entity and the Borrowing entity in the ER diagram). (2)
- c) [PLACEHOLDER FIGURE] (Explain how the system can ensure data integrity for the Librarian entity.). (2)
- d) List three types of constraints that can be used to enforce data integrity in the cataloging system. (3)
- e) Describe the advantages of using a relational model for the University Library's cataloging system. (1)
- f) [PLACEHOLDER FIGURE] (Map the ER diagram to a relational schema, including the primary and foreign keys.). (2)
- g) Explain how the system can be optimized for efficient querying and retrieval of data. (2)